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Calibration of Production Transient Classifiers: A Benchmark Study

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ABSTRACT

The Vera C. Rubin Observatory's Legacy Survey of Space and Time will generate $\sim 10^7$ transient alerts per night against a global spectroscopic follow-up capacity of $\sim 10^5$ observations per year. Automated alert brokers address this mismatch by outputting class probability vectors, which astronomers use to prioritise follow-up observations. These probabilities are implicitly treated as reliable confidence estimates, yet to our knowledge no published work has evaluated the calibration of production transient classifiers against independent spectroscopic ground truth.

We present a systematic calibration audit of production astronomical transient classifiers, evaluating ALeRCE's light curve classifier and two Fink classifiers — a Random Forest and SuperNNova binary discriminator — on 1,606 spectroscopically confirmed objects from the ZTF Bright Transient Survey. We compute Expected Calibration Error (ECE) with bootstrap confidence intervals, report per-class calibration under three explicit definitions, and pair ECE with ranking metrics to distinguish repairable miscalibration from uninformative scores.

ALeRCE's classifier exhibits systematic underconfidence (ECE = 0.259; 95% CI: $[0.237, 0.280]$), with predicted probabilities consistently below empirical accuracies across all confidence bins. We trace this to directly observable confidence compression: no object receives top-1 confidence above 0.88, and among correctly classified SNIa the maximum is 0.72 against 0.83 accuracy. The compression produces a pronounced class-conditional confidence deficit for the best-classified types (SNIa, SNIbc), while under the standard one-vs-rest definition miscalibration is more uniform across classes. TDE achieves classification accuracy of 0.000 because TDE is absent from the light curve classifier's output taxonomy. Temperature scaling fitted and evaluated with held-out validation ($T = 0.361 \pm 0.011$, 5-fold cross-validation) reduces aggregate ECE by 94% to 0.015, requiring no model retraining.

The Fink binary scores are severely miscalibrated on this sample (ECE = 0.464 for RF, 0.304 for SuperNNova) and, more fundamentally, carry no measurable ranking signal for SNIa identification (ROC-AUC of 0.502 and 0.525): no monotone post-hoc correction can recover informative posteriors, and restricting the RF to early-phase proxy subsets does not improve it.

These results demonstrate that miscalibration is systematic, broker-dependent, and consequential for confidence-based follow-up prioritisation. In this bright spectroscopic sample, ALeRCE's probabilities are reliably correctable post hoc, while Fink's available binary scores are unsuitable as general-purpose SNIa posteriors. We argue that calibration metrics should be reported alongside accuracy — and alongside ranking metrics — for production astronomical classifiers, and that calibrated ALeRCE outputs are a promising starting point for deferral-aware classification frameworks, pending validation on fainter LSST-like samples.

Keywords: astronomical databases — surveys — supernovae: general — methods: statistical

1. INTRODUCTION

The Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST; Ivezić et al. 2019) will generate approximately 10^7 transient alerts per night beginning in 2025. Global spectroscopic follow-up capacity — the essential confirmation mechanism for transient classification — is fixed at roughly 10^5 observations per year (Hambly et al. 2022). This mismatch of four orders of magnitude transforms transient astronomy from a data-limited to a decision-limited discipline: the central challenge is no longer detecting transients but determining which ones warrant the scarce resource of spectroscopic confirmation.

Automated alert brokers — ALeRCE (Förster et al. 2021), Fink (Möller et al. 2021), ANTARES (Narayan et al. 2018), and Lasair (Smith et al. 2019) — address this challenge by classifying transients from photometric light curves and outputting class probability vectors. These probabilities are used by astronomers and automated pipelines to prioritise follow-up observations. Implicit in this usage is an assumption that the probabilities are meaningful confidence estimates — that a classifier outputting $P(\text{SNIa}) = 0.9$ is correct approximately 90% of the time.

To our knowledge, this assumption has not been tested: we are not aware of published work evaluating the calibration of production transient classifiers using independent spectroscopic ground truth (Section 2). Reliability diagrams and Expected Calibration Error (ECE; Naeini et al. 2015) — standard diagnostic tools in machine learning — are largely absent from the astronomical broker literature. If production classifiers are systematically miscalibrated, the confidence-based prioritisation decisions made by astronomers and automated systems are built on unreliable foundations.

In this paper we conduct a systematic calibration audit of production transient classifiers, evaluating ALeRCE's light curve classifier and two Fink classifiers on 1,606 spectroscopically confirmed objects from the ZTF Bright Transient Survey (Fremling et al. 2020; Perley et al. 2020). Our principal findings are:

1. ALeRCE's classifier exhibits systematic underconfidence (ECE = 0.259), not the overconfidence typical of deep learning models, and we directly observe its cause: the classifier's confidence range is compressed (global maximum 0.88; maximum 0.72 among correctly classified SNIa).

2. The compression produces an inverse relationship between accuracy and the *class-conditional confidence deficit*: true members of the best-classified types (SNIa, SNIbc; accuracy > 0.83) receive confidences far below their achieved accuracy. Under the standard one-vs-rest classwise definition, miscalibration is more uniform across classes; we report both explicitly.

3. Temperature scaling with a single scalar parameter, fitted and evaluated on held-out data ($T = 0.361 \pm 0.011$, 5-fold CV), reduces ALeRCE's ECE by 94% to 0.015, with no model retraining required.

4. TDE, a high-priority LSST science target, achieves classification accuracy of 0.000 because TDE is absent from the light curve classifier's output taxonomy.

5. Fink's binary SNIa scores are not only severely miscalibrated on this sample (ECE = 0.464 for RF, 0.304 for SuperNNova) but carry no measurable ranking signal (ROC-AUC ≈ 0.5), so no monotone post-hoc method can recover informative posteriors from them.

These findings have immediate implications for spectroscopic follow-up prioritisation and motivate the deferral-aware classification framework we develop in subsequent work. Section 2 reviews related work on broker classification and calibration methods. Section 3 describes our dataset. Section 4 details our calibration methodology. Section 5 presents results. Section 6 discusses implications and limitations.

2. RELATED WORK

2.1. *Astronomical Alert Brokers*

The ZTF alert stream has prompted the development of several automated classification brokers. ALeRCE (Förster et al. 2021) operates a two-stage pipeline: a stamp classifier for early real/bogus filtering and a light curve classifier (Sánchez-Sáez et al. 2021; Pavez-Herrera et al. 2025) for multi-class transient classification using Random Forest on extracted photometric features. Fink (Möller et al. 2021) provides a modular science portal with multiple classifiers operating independently, including the SuperNNova-based binary classifier (Möller & de Boissière 2020) and a Random Forest SN Ia discriminator. ANTARES (Narayan et al. 2018) focuses on real-time annotation and filtering. Lasair (Smith et al. 2019) provides alert stream access with community-contributed classifiers.

All major brokers provide classification probabilities as outputs. None, to our knowledge, publish calibration metrics such as ECE or reliability diagrams for their production classifiers. This gap motivates the present work.

2.2. *Calibration in Machine Learning*

The calibration of neural network classifiers has received substantial attention since Guo et al. (2017) demonstrated that modern deep networks are systematically overconfident and that temperature scaling provides an effective post-hoc remedy. Subsequent work has examined calibration under distribution shift (Ovadia et al. 2019), for binary classifiers (Kull & Flach 2017), and in the context of selective prediction (Geifman & El-Yaniv 2017).

Calibration of Random Forest classifiers is less studied but well understood at a mechanistic level. Niculescu-Mizil & Caruana (2005) demonstrated that RF produces systematically distorted probability estimates — underconfident for well-discriminated classes and poorly calibrated near class boundaries — and that Platt scaling (Platt 1999) and isotonic regression provide effective corrections. Zadrozny & Elkan (2002) introduced isotonic regression as a nonparametric post-hoc calibration method applicable to any classifier; Kull & Flach (2017) proposed beta calibration for binary scores, and Kull et al. (2019) extended matrix/Dirichlet scaling to the multiclass setting. The sensitivity of ECE itself to bin construction is analysed by Nixon et al. (2019) and Roelofs et al. (2022), motivating the adaptive-binning robustness checks we perform in Section 5.7.

In astronomy, calibration has been studied in the context of photometric redshift estimation (Bordoloi et al. 2010; Schmidt et al. 2020), where reliable uncertainty quantification directly affects cosmological inference. Calibration of transient classifiers has not, to our knowledge, been systematically evaluated on real spectroscopic samples prior to this work.

2.3. *Uncertainty Quantification for Transient Classification*

Uncertainty quantification for astronomical transient classification has been explored primarily through Bayesian and ensemble approaches. Möller et al. (2016) applied Bayesian neural networks to supernova photometric classification, obtaining uncertainty estimates alongside class predictions. Villar et al. (2019) used deep ensembles for the PLAsTiCC challenge, demonstrating that ensemble disagreement correlates with classification difficulty.

These works focus on model uncertainty (epistemic uncertainty from limited training data) rather than calibration (the alignment between predicted confidence and empirical accuracy). The two concepts are related but distinct: a well-calibrated classifier need not have low uncertainty, and a classifier with well-quantified uncertainty need not be calibrated. Our focus on ECE and reliability diagrams addresses the calibration question directly, complementing prior uncertainty quantification work.

2.4. *Learning to Defer*

Learning-to-defer (L2D; Madras et al. 2018) is a framework for training joint classifier-deferral systems that learn when to route decisions to a human expert. Mozannar & Sontag (2020) extended L2D to handle consistent surrogate losses, and Verma et al. (2022) developed calibrated L2D systems that provide reliable confidence estimates alongside deferral decisions.

L2D has been applied in medical imaging (Tschandl et al. 2019), content moderation, and autonomous driving, where human-AI collaboration under resource constraints is operationally critical. To our knowledge, L2D has not been applied to astronomical transient classification. The present calibration study establishes the empirical foundation for this application: demonstrating that production classifiers are miscalibrated motivates the need for explicit deferral mechanisms, and providing post-hoc calibration methods identifies the broker most suitable for downstream L2D integration.

3. DATA

3.1. *The ZTF Bright Transient Survey*

Our dataset is drawn from the Zwicky Transient Facility Bright Transient Survey (BTS; Fremling et al. 2020; Perley et al. 2020), a magnitude-limited ($m < 18.5$) spectroscopic completeness survey operating since 2018. The BTS provides spectroscopic classifications for ZTF transients, making it the largest homogeneous sample of spectroscopically confirmed extragalactic transients currently available. We use spectroscopic classifications as ground truth labels throughout this work.

3.2. *Sample Construction*

We queried the ALeRCE broker API (Förster et al. 2021) for classification probabilities corresponding to all BTS objects with confirmed spectroscopic types on 22 February 2026. ALeRCE's light curve classifier (Sánchez-Sáez et al. 2021) outputs probabilities across 15 classes derived from features of the ZTF alert photometric time series. For each object we retrieved the most recent classification available at query time. Our initial collection did not record the `classifier_version` tag returned alongside each probability vector; we later re-queried the API specifically to recover it (Section 5.2) and found two distinct versions active across our collection window, `hierarchical_rf_1.1.0` and `lc_classifier_1.1.13`, for 1,601 of 1,606 objects (5 API failures on re-query). We provide the full list of ZTF object identifiers, retrieved probability vectors, and recovered version tags as a machine-readable supplement to this paper.

We map the 11 BTS spectroscopic types to the 5 ALeRCE transient classes most relevant to extragalactic time-domain science: Type Ia supernovae (SNIa), Type II supernovae (SNII), stripped-envelope supernovae (SNIbc), superluminous supernovae (SLSN), and tidal disruption events (TDE). The full mapping is given in Table 1.

Table 1. BTS spectroscopic type to ALeRCE class mapping.

BTS Types	ALeRCE Class
SN Ia, SN Ia-91T, SN Ia-91bg, SN Ia-pec	SNIa
SN II, SN IIn, SN IIb, SN IIP	SNII
SN Ib, SN Ic, SN Ib/c, SN Ic-BL	SNIbc
SLSN-I	SLSN
TDE	TDE

Of 1,987 BTS objects queried, 1,606 returned valid ALeRCE classifications (80.1% success rate). Failed queries reflect objects observed before ALeRCE's alert ingestion began or with insufficient photometric history for classification. The final sample comprises 757 SNIa, 560 SNII, 204 SNIbc, 55 SLSN, and 30 TDE.

To avoid class imbalance biasing our rarest-class estimates, we collected all available BTS objects in the SLSN and TDE categories rather than applying a fixed per-class cap. This yields a sample in which common classes (SNIa, SNII) are well-represented while preserving the full available statistics for scientifically rare but high-priority classes.

3.3. *Fink Broker Data*

We additionally queried the Fink broker (Möller et al. 2021) for the same set of ZTF objects via the Fink Portal API on 24 February 2026, using HTTP POST requests to `api.fink-portal.org/api/v1/objects`. Fink provides two binary classifiers relevant to our analysis. `snn_snia_vs_nonia` is based on the SuperNNova deep learning classifier (Möller & de Boissière 2020) and outputs a general $P(\mathrm{SNIa})$ score. `rf_snia_vs_nonia` is a Random Forest classifier documented as targeting *rising* SN Ia candidates; its scores are designed for early-phase, pre-maximum alerts rather than general-purpose SN Ia posteriors. We include both scores in our analysis but explicitly flag this distinction when interpreting the RF results in Sections 5.6 and 6.4.

For each object we retrieved the most recent alert-level score available in the Fink database. We distinguish three outcomes: object found with scores (collected), object absent from Fink (not in ingestion window, treated as missing data), and API error (excluded). Absent objects are not coerced to zero — they are excluded from the Fink analysis entirely.

Of 1,606 objects queried, 1,368 (85.2%) were present in the Fink database, with 237 objects absent due to falling outside Fink's alert ingestion window. Both classifiers returned scores for all 1,368 matched objects, yielding a comparative subsample of 644 SNIa, 480 SNII, 166 SNIbc, 50 SLSN, and 28 TDE.

3.4. *Ground Truth and Selection Effects*

We emphasise that our ground truth labels derive from independent spectroscopic observations, not from the broker classifications being evaluated. This avoids the circular validation present in studies that use broker-assigned labels to assess broker performance.

The BTS magnitude limit ($m < 18.5$) introduces a selection bias toward brighter, lower-redshift transients. Our sample therefore represents the regime in which spectroscopic follow-up is most feasible, which is precisely the regime most relevant to the deferral decisions this work motivates. We do not claim our calibration results generalise to fainter or higher-redshift populations. As the BTS selects transients for spectroscopic follow-up partly on the basis of brightness and observability, our sample likely over-represents well-behaved, unambiguous examples of each class. Calibration performance on the fainter, more ambiguous objects that will dominate LSST alert streams may differ; our results should be interpreted as an upper bound on broker performance in the bright regime.

4. METHODS

4.1. *Expected Calibration Error*

A classifier is said to be *calibrated* if its predicted confidence matches its empirical accuracy. Formally, for a classifier outputting confidence $\hat{p}$ and true label y:

$$P(y = \hat{y} \mid \hat{p} = p) = p \quad \forall\, p \in [0, 1] \tag{1}$$

where $\hat{y}$ is the predicted class. We measure calibration using Expected Calibration Error (ECE; Naeini et al. 2015), which approximates the expectation in Equation 1 by partitioning predictions into M equally spaced confidence bins:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right| \tag{2}$$

where B_m is the set of samples whose top-1 confidence falls in bin m, n is the total number of samples, $\mathrm{acc}(B_m)$ is the fraction of correct predictions in bin m, and $\mathrm{conf}(B_m)$ is the mean confidence in bin m. We use $M = 10$ bins throughout. A perfectly calibrated classifier has ECE $= 0$; following common practice we use 0.05 as a practical reference level for consequential miscalibration (Guo et al. 2017), while noting this is a convention, not a formal significance boundary.

For ALeRCE's 15-class classifier, we compute ECE using the top-1 predicted class and its associated probability across all 15 output classes. For Fink's binary classifiers, which output a scalar $P(\text{SNIa})$, we apply a binary variant of Equation 2 where $\text{acc}(B_m)$ is the observed SNIa fraction within bin m.

4.1.1. *Per-class calibration: three definitions*

"Per-class calibration" is ambiguous in the multiclass setting, and the choice of definition materially affects conclusions (Section 5.3). We therefore compute and report three distinct quantities explicitly:

1. **True-class-conditioned top-label ECE**: for class k, Equation 2 restricted to objects whose *spectroscopic* label is k. This measures a class-conditional confidence deficit — how appropriately the classifier weights objects that truly are class k — which is the operationally relevant quantity when asking whether genuine members of a class will clear a confidence-based follow-up threshold. It is *not* calibration in the standard conditional sense.
2. **Predicted-class-stratified top-label ECE**: Equation 2 restricted to objects whose *predicted* top-1 class is k. This asks whether confidence matches precision within each predicted-class stream.
3. **One-vs-rest classwise ECE**: over *all* objects, binning the class-k probability p_k against the empirical frequency of true class k — the standard classwise calibration condition $P(y = k \mid \hat{p}_k = p) = p$.

4.1.2. *Ranking metrics*

ECE alone cannot distinguish a classifier that ranks objects usefully but reports distorted probabilities (repairable post hoc) from one with no discriminative signal (not repairable by any monotone transformation, since monotone recalibration preserves ranking). For the binary Fink classifiers we therefore also report ROC–AUC and average precision (PR–AUC), alongside Brier score. A score with AUC ≈ 0.5 can always be made to *appear* calibrated — a constant prediction at the base rate has ECE ≈ 0 — while carrying no information; reporting ECE together with ranking metrics guards against this failure mode.

4.1.3. *Robustness to bin construction*

Equation 2's equal-width bins are known to be sensitive to how probability mass is distributed across the confidence range (Nixon et al. 2019; Roelofs et al. 2022): a classifier whose scores concentrate near one end of $[0, 1]$ — Fink's `rf_snia_vs_nonia` has mean score 0.011, putting roughly 90% of objects in the first of 10 bins — leaves the remaining bins populated by very few points, so the resulting ECE can be more sensitive to noise or bin placement than a well-populated case. As a robustness check, we additionally compute each headline ECE using equal-*mass* (adaptive) bins, i.e. quantile bins of the confidence/score distribution that guarantee $\approx n/10$ samples per bin regardless of skew. If equal-width and equal-mass binning agree, the reported miscalibration is a property of the classifier and not an artefact of bin construction.

4.2. *Uncertainty Quantification via Bootstrap*

Point estimates of ECE carry substantial sampling uncertainty, particularly for small classes such as SLSN ($n = 55$) and TDE ($n = 30$). We report 95% confidence intervals on all ECE estimates using the nonparametric bootstrap (Efron 1979): we resample the evaluation set with replacement $B = 2000$ times, compute ECE on each resample, and report the 2.5th and 97.5th percentiles of the resulting distribution. All bootstrap procedures use a fixed random seed (seed = 42) for reproducibility.

4.3. *Reliability Diagrams*

We visualise calibration using reliability diagrams (DeGroot & Fienberg 1983), which plot observed accuracy against mean predicted confidence for each bin. A perfectly calibrated classifier produces points lying on the diagonal. Points above the diagonal indicate underconfidence (predicted probability lower than empirical accuracy); points below indicate overconfidence. We include sample-count histograms alongside each diagram to indicate where estimates are statistically reliable.

4.4. *Temperature Scaling*

Temperature scaling (Guo et al. 2017) is a post-hoc calibration method that applies a single scalar parameter $T > 0$ to the pre-softmax logits of a classifier. For a classifier outputting probability vector $\mathbf{p}$, the calibrated probabilities are:

$$\hat{\mathbf{p}}_T = \text{softmax}\left(\frac{\log \mathbf{p}}{T}\right) \tag{3}$$

Values of $T < 1$ increase confidence (sharpening the distribution); values of $T > 1$ decrease confidence (flattening toward uniform). The optimal T is found by minimising ECE on the evaluation set using bounded scalar optimisation over $T \in [0.1, 10.0]$.

For Fink's binary classifiers, we apply an equivalent logit-space temperature scaling:

$$\hat{p}_T = \sigma\left(\frac{\text{logit}(p)}{T}\right) \tag{4}$$

where σ is the sigmoid function and $\text{logit}(p) = \log(p/(1-p))$.

Fitting T on the same data used to evaluate it yields an optimistic estimate. Our primary results therefore use held-out evaluation throughout: 5-fold stratified cross-validation (T fitted on four folds by NLL minimisation, ECE evaluated on the held-out fold, pooled across folds), corroborated by an independent 60/40 calibration/test split. We additionally report the same-set fit as an explicit *oracle upper bound*, labelled as such, for comparison with the held-out numbers.

For the Fink classifiers we also fit isotonic regression (Zadrozny & Elkan 2002) under the same 5-fold cross-validation — a substantially more flexible monotone calibrator than scalar temperature — so that failure of temperature scaling alone is not over-interpreted as failure of all post-hoc methods. Isotonic results must be read jointly with the ranking metrics of Section 4.1.2: a calibrator applied to an uninformative score can achieve low ECE by collapsing predictions toward the base rate.

4.5. *Probability Renormalisation*

ALeRCE's classifier outputs probabilities across 15 classes, of which we evaluate 4 (excluding TDE, which is absent from ALeRCE's taxonomy). To assess whether apparent miscalibration arises from probability dilution across unevaluated classes, we compute a renormalised ECE by projecting the 15-class probability vector onto the 4 evaluated classes and renormalising to sum to unity. The difference between 15-class and renormalised ECE quantifies the contribution of probability dilution to the observed miscalibration.

5. RESULTS

5.1. *ALeRCE Aggregate Calibration*

Figure 1 shows the reliability diagram for ALeRCE's light curve classifier evaluated on 1,576 spectroscopically confirmed objects (30 TDE excluded; see Section 5.4). The classifier exhibits systematic underconfidence across all confidence bins: every point lies above the perfect calibration diagonal, indicating that predicted probabilities are consistently lower than empirical accuracies.

The aggregate ECE is 0.259 (95% CI: $[0.237, 0.280]$), well above the 0.05 threshold for practically significant miscalibration. The overall accuracy is 0.709 while the mean top-1 confidence is 0.451, a gap of +0.258 that is consistent with the ECE estimate. This gap is directional and consistent — not a consequence of random binning artefacts — indicating a systematic mechanistic suppression of classifier confidence.

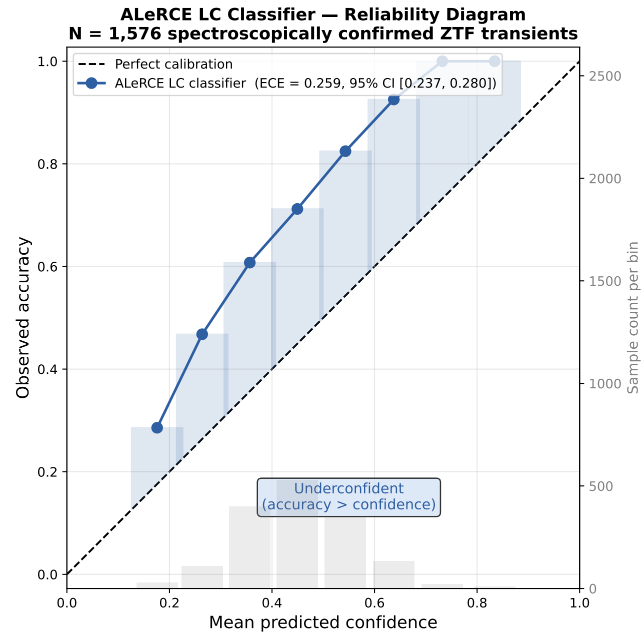

Figure 1. Reliability diagram for ALeRCE's light curve classifier evaluated on 1,576 spectroscopically confirmed ZTF transients (30 TDE excluded). Every point lies above the perfect calibration diagonal, indicating systematic underconfidence across all confidence bins. The histogram shows sample counts per bin. ECE = 0.259 (95% CI: $[0.237, 0.280]$).

To assess whether this miscalibration arises from probability dilution across ALeRCE's 15-class output space, we renormalise the probability vectors onto the 4 evaluated classes and recompute ECE. The renormalised ECE is 0.251 (95% CI: $[0.230, 0.272]$), a difference of 0.008 from the 15-class estimate. Probability dilution accounts for only 3% of the observed miscalibration; the remainder is intrinsic to the classifier.

5.2. *Stability Across Classifier Versions*

Our sample was collected by querying each object's most recent classification at the time of retrieval (Section 3.2), which spans whatever `lc_classifier` version was live in production when each query was made. Pooling predictions across versions is only valid if calibration is stable across them; we test this directly rather than assuming it. We re-queried the ALeRCE API for the `classifier_version` tag attached to each object (recovered for 1,601 of 1,606 objects; 5 API failures) and found two distinct versions in production over our collection window: `hierarchical_rf_1.1.0` ($n = 1346$) and a newer `lc_classifier_1.1.13` ($n = 225$) — not

Table 2. ALeRCE 15-class top-1 ECE and fitted tempera[illegible] by classifier version (TDE excluded; $n = 1571$ of 1576 [illegible] objects had no recoverable version tag).

Version	N	ECE (95% CI)	T
`hierarchical_rf_1.1.0`	1346	0.258 [0.235, 0.281]	0.382
`lc_classifier_1.1.13`	225	0.260 [0.200, 0.317]	0.326
Pooled (this subset)	1571	0.258 [0.236, 0.278]	0.355

Note—The pooled row uses the 1,571-object subset with a recovered version tag; it is consistent with the full 1,576-object headline estimate (ECE = 0.259) as expected from dropping only 5 objects.

Table 3. ALeRCE per-class calibration under the three definitions of Section 4.1.1, with bootstrap 95% CIs (2,000 resamples). N is the number of objects entering each estimate: true class members (1), predicted class members (2), or all objects (3).

Class	Acc	(1) true-class	(2) pred.-class	(3) one-vs-rest
SNIa	0.83	0.376 [0.352, 0.401]	0.392 [0.367, 0.415]	0.204 [0.187, 0.224]
SNII	0.49	0.099 [0.066, 0.141]	0.461 [0.425, 0.495]	0.143 [0.129, 0.164]
SNIbc	0.83	0.360 [0.315, 0.405]	0.080 [0.053, 0.133]	0.158 [0.146, 0.171]
SLSN	0.78	0.225 [0.181, 0.319]	0.284 [0.245, 0.342]	0.134 [0.126, 0.142]

Note—N per column: (1) 757/560/204/55; (2) 740/315/343/149; (3) 1576 for all. SLSN estimates under definitions (1)–(2) carry relative CI widths of ∼50% and should be read as suggestive. TDE is excluded (Section 5.4): its 0.000 accuracy reflects a taxonomy gap, not a calibration estimate.

merely a version-number increment but a differently-named classifier product, indicating a genuine model transition rather than a minor patch.

Table 2 shows 15-class top-1 ECE — the same quantity as the headline estimate — and the fitted temperature separately for each version (TDE and the 5 unmatched objects excluded, $n = 1571$). The two versions' ECEs are nearly identical (0.258 vs. 0.260) with strongly overlapping 95% CIs, and both fitted temperatures lie in the same sharpening regime ($T = 0.38$ vs. 0.33; the smaller version's $n = 225$ makes its T less precisely determined). We find no evidence that pooling across versions is inflating or distorting the aggregate ECE or the temperature correction of Section 5.5.

Two caveats. First, the version tags were recovered by a later re-query of the API: if the broker recomputed an object's classification between our original retrieval and the re-query, the tag may describe a newer output than the probability vector we analyse. The check is therefore best-available, not perfect, version attribution. Second, this check covers ALeRCE only; we could not perform the equivalent test for Fink.

5.3. *ALeRCE Per-Class Calibration*

Figure 2 shows per-class reliability diagrams for the four ALeRCE-classified transient types. Table 3 reports per-class calibration under all three definitions of Section 4.1.1; the choice of definition matters, so we describe each in turn.

Under the true-class-conditioned definition (1) — the operationally relevant one for asking whether genuine members of a class will clear a confidence threshold — there is a pronounced inverse relationship between classification accuracy and confidence deficit. SNIa (accuracy 0.831) shows ECE = 0.377 (95% CI: $[0.352, 0.401]$) and SNIbc (accuracy 0.828) shows 0.362 $[0.315, 0.405]$: both are classified correctly more than 80% of the time, yet receive confidences far below that level. SNII (accuracy 0.495) shows the smallest deficit (0.103 $[0.066, 0.141]$).

This inversion is, however, **definition-dependent**, and we flag this explicitly. Under predicted-class stratification (definition 2) the ordering changes substantially: objects *predicted* SNII carry the largest gap (ECE = 0.461 $[0.425, 0.495]$ — the classifier's precision on predicted-SNII objects far exceeds its stated confidence), while predicted-SNIbc is the best calibrated (0.080). Under the standard one-vs-rest classwise definition (3), all four classes are significantly miscalibrated at a more uniform level (0.13–0.20), with SNIa worst (0.204). Two conclusions are robust across all three definitions: the direction is underconfidence throughout, and every class exceeds the 0.05 reference level. The specific claim "the best-classified classes are the worst-calibrated" holds for the class-conditional confidence deficit (definition 1), not for classwise calibration in the standard sense.

The confidence deficit for well-classified classes is consistent with a known property of Random Forest probability estimates: RF outputs are averages of tree votes, and under class imbalance this averaging compresses predicted probabilities away from the extremes (Niculescu-Mizil & Caruana 2005; Zadrozny & Elkan 2002). We can demonstrate the compression directly in our sample rather than leaving it as a hypothesis: among the 629 *correctly classified* SNIa, the maximum top-1 confidence assigned to any object is 0.72 and the median is 0.47 — against an achieved accuracy of 0.83. Across the full sample, only 0.6% of objects receive confidence above 0.8, none above 0.9, and the global

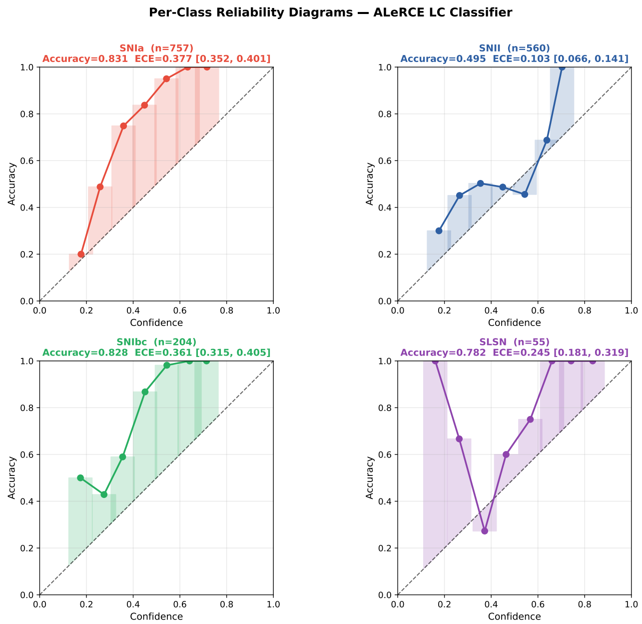

Figure 2. Per-class reliability diagrams for ALeRCE's light curve classifier. SNIa and SNIbc (accuracy > 0.83) show severe underconfidence (ECE > 0.36). SNII (accuracy 0.495) is the best-calibrated class (ECE $= 0.103$). SLSN shows wider confidence intervals reflecting the smaller sample size ($n = 55$).

maximum is 0.88. The output range is truncated well below the level required to match the accuracy of the best-classified types, which mechanically forces a confidence deficit for any class with accuracy above $\sim$0.72. Whether this compression originates specifically from inverse-frequency class weighting in training would require access to ALeRCE's training configuration; the compression itself, however, is directly observed.

5.4. *A Taxonomy Gap in the Deployed Classifier*

TDE achieves a classification accuracy of 0.000 on our 30-object sample. This is not a calibration problem and not a shortcoming of the model on a task it was built for: TDE is simply *absent* from the deployed classifier's output space. The Sánchez-Sáez et al. (2021) light curve classifier resolves four transient classes — SNIa, SNIbc, SNII, SLSN — and cannot emit a TDE probability by construction, so it assigns all mass to other classes for every TDE. We confirmed this holds for both production versions in our sample by re-querying the live API: `hierarchical_rf_1.1.0` and `lc_classifier_1.1.13` each return the same 15-class taxonomy with no TDE class.

We stress that this is a statement about the *deployed* classifier, not about ALeRCE as a project. A TDE-aware extension of the light curve classifier has since been developed (Pavez-Herrera et al. 2025), adding TDE as a subclass and reaching $\sim$91% TDE recall on a labelled set. That version was not serving the alert stream during our collection window, so the operational point stands in a sharper, narrower form: the capability to classify TDEs exists in the literature but had not yet reached the classifier whose probability vectors actually drive follow-up decisions. TDEs are a high-priority LSST science case (Gezari 2021), and an object that is confidently mislabelled as a supernova — rather than flagged as uncertain — is precisely the silent failure mode that confidence-based prioritisation is meant to avoid. We exclude TDE from all calibration analyses (they carry no ECE by construction) but retain the observation as motivation for verifying taxonomy coverage, and for out-of-taxonomy novelty detection, before LSST first light.

5.5. *Temperature Scaling*

We apply temperature scaling (Section 4.4) to ALeRCE's classifier outputs. Under 5-fold cross-validation the fitted temperature is stable across folds ($T = 0.361 \pm 0.011$), substantially below unity, confirming that the classifier requires confidence *sharpening* — the opposite of the overconfidence correction for which temperature scaling is most commonly applied (Guo et al. 2017).

Our primary, held-out result: pooling the held-out folds gives ECE $= 0.015$ (95% CI: $[0.017, 0.045]$ across bootstrap resamples of the pooled folds), a 94% reduction from the pre-calibration 0.259; Brier score improves from 0.127 to 0.103 and NLL from 0.938 to 0.787. The independent 60/40 calibration/test split corroborates this: T fitted on 945 objects yields ECE $= 0.024$ on the 631 held-out objects. For comparison, the same-set oracle fit ($T = 0.359$ on all 1,576 objects) gives ECE $= 0.031$ — the held-out numbers are not meaningfully worse than the oracle bound, indicating the correction generalises rather than overfitting the evaluation set.

Figure 3 shows the reliability diagram before and after scaling. Mean confidence increases from 0.451 to 0.712, closely matching the empirical accuracy of 0.709.

Table 4 shows per-class ECE before and after scaling. Temperature scaling substantially improves SNIa ($0.377 \rightarrow 0.114$) and SNIbc ($0.360 \rightarrow 0.108$), classes with $n \geq 204$ and correspondingly tight bootstrap CIs (Table 3). SNII *worsens* from 0.099 to 0.189. The apparent "moderate improvement" for SLSN ($0.225 \rightarrow 0.140$) should be read more cautiously: SLSN's pre-scaling ECE alone carries a 95% CI of $[0.181, 0.319]$ on $n = 55$ objects (Section 5.3), a range wide enough that the point estimate of the post-scaling value is only

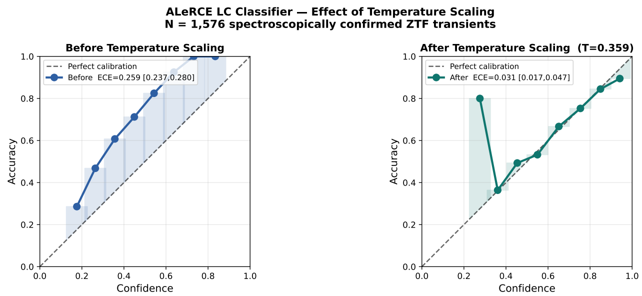

Figure 3. ALeRCE reliability diagrams before (left) and after (right) temperature scaling with $T = 0.359$. ECE reduces from 0.259 to 0.031 — an 88% improvement. The residual dip near confidence 0.3 in the right panel reflects SNII overcorrection (see Section 6.3).

Table 4. ALeRCE per-class ECE (true-class-conditioned; definition 1 of Section 4.1.1) before and after 5-fold cross-validated temperature scaling ($T = 0.361 \pm 0.011$).

Class	N	ECE (before)	ECE (after)	ΔECE
SNIa	757	0.377	0.114	−0.263
SNII	560	0.099	0.189	+0.090
SNIbc	204	0.360	0.108	−0.252
SLSN	55	0.225	0.140	−0.085

NOTE—Positive ΔECE indicates worsening after temperature scaling. SNII is the only class for which global temperature scaling is counterproductive.

weakly constrained; we report it for completeness but do not treat the specific 0.085 improvement as a precise measurement.

This result is mechanistically interpretable. SNII was the only well-calibrated class before scaling: the classifier's uncertainty was already proportionate to its error rate. A global $T < 1$ uniformly increases confidence across all classes; for SNII this produces overconfidence where none previously existed. A single global temperature parameter cannot simultaneously optimise calibration for classes with heterogeneous miscalibration characteristics. Class-specific temperature parameters would be required for a complete fix.

5.6. *Fink Binary Classifiers*

Figure 4 shows reliability diagrams for Fink's two binary classifiers evaluated on 1,368 objects. Both exhibit severe miscalibration, but with qualitatively different failure modes.

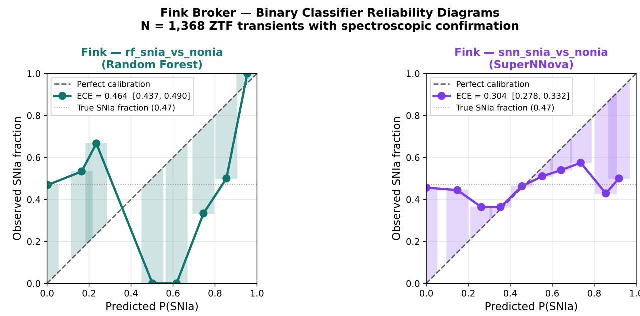

Figure 4. Reliability diagrams for Fink's two binary classifiers evaluated on 1,368 ZTF transients. Left: Random Forest (`rf_snia_vs_nonia`), ECE = 0.464. The flat curve indicates no discriminative power for SNIa identification. Right: SuperNNova (`snn_snia_vs_nonia`), ECE = 0.304. Weak discriminative signal persists but is severely suppressed.

The Random Forest classifier (`rf_snia_vs_nonia`) assigns near-zero SNIa probability to virtually all objects: mean score 0.011 against a true SNIa fraction of 0.471. The reliability curve is essentially flat at the base rate regardless of predicted confidence, indicating the classifier has no discriminative power for SNIa identification in this sample (ECE = 0.464, 95% CI: $[0.437, 0.490]$).

SuperNNova (`snn_snia_vs_nonia`) shows qualitatively better behaviour: mean score 0.222 against true fraction 0.471, with a reliability curve that rises weakly with confidence (ECE = 0.304, 95% CI: $[0.278, 0.332]$). The reliability curve's weak rise suggests marginal discriminative signal; the ranking metrics below quantify how marginal (ROC-AUC = 0.525).

Temperature scaling applied to both Fink classifiers hits the upper optimisation bound ($T \approx 10$) in both cases, achieving only modest ECE reductions: 24% for RF ($0.464 \to 0.353$) and 25% for SuperNNova ($0.304 \to 0.229$). Failure of scalar temperature scaling alone, however, does not establish that no post-hoc method can help. We therefore add two diagnostics (Table 5).

First, ranking metrics. On this sample both scores carry essentially no discriminative signal for SNIa identification: ROC-AUC = 0.502 for RF and 0.525 for SuperNNova (0.5 = chance), with average precision (0.476, 0.493) indistinguishable from the SNIa base rate of 0.471. This is the decisive measurement: monotone recalibration preserves ranking, so with AUC $\approx$ 0.5 *no* monotone post-hoc transformation — temperature, Platt, isotonic, beta — can produce scores that are simultaneously calibrated and informative on this sample.

Second, isotonic regression under 5-fold cross-validation *does* achieve low ECE (0.014 for RF, 0.008 for SNN) — but this is exactly the calibrated-yet-uninformative failure mode anticipated in Section 4.1.2: with no ranking signal to preserve, the isotonic fit col-

Table 5. Fink classifier diagnostics beyond ECE. Isotonic ECE is 5-fold cross-validated. AUC = 0.5 indicates no ranking signal; average precision (PR-AUC) at the base rate (0.471) likewise.

Classifier	Subset	ECE	ECE (isotonic)	ROC-AUC	PR-AUC
RF	full ($n = 1368$)	0.464	0.014	0.502	0.476
RF	`ndethist` ≤ 5	0.470	0.002	0.499	0.475
RF	`ndethist` ≤ 3	0.464	0.000	0.500	0.464
SNN	full ($n = 1368$)	0.300	0.008	0.525	0.493
SNN	`ndethist` ≤ 5	0.312	0.011	0.536	0.500
SNN	`ndethist` ≤ 3	0.325	0.011	0.540	0.492

NOTE—Low isotonic ECE here reflects collapse toward the base rate (calibrated but uninformative), not recovery of useful posteriors — see the AUC columns.

Table 6. ECE under equal-width vs. equal-mass (adaptive) binning, 10 bins, 2,000-resample bootstrap CIs.

Classifier	N	Equal-width ECE	Equal-mass ECE
ALeRCE aggregate	1576	0.259 [0.237, 0.280]	0.259 [0.237, 0.280]
Fink RF	1368	0.464 [0.437, 0.490]	0.460 [0.433, 0.485]
Fink SNN	1368	0.300 [0.278, 0.332]	0.302 [0.278, 0.332]

NOTE—Equal-mass bins guarantee $\approx n/10$ samples per bin regardless of score skew. Agreement with equal-width ECE (largest relative change: 1.0%, Fink RF) indicates the reported miscalibration is not a binning artefact. Entries here are point estimates of Equation 2; headline values quoted elsewhere (e.g. SNN 0.304) are bootstrap means, which differ from the point estimate (0.300) at the third decimal.

lapses predictions toward the base rate, which is trivially calibrated and carries no per-object information. We report this as a caution against reading ECE in isolation, not as a repair of the Fink scores.

Finally, because the RF classifier is documented as targeting *rising*, early-phase candidates, we recompute all metrics on early-phase proxy subsets (alerts with ≤ 5 and ≤ 3 prior detections, `ndethist`; 75% and 62% of the sample respectively). Neither ECE nor AUC improves (Table 5) — within the limits of this proxy, the lack of discriminative signal is not explained by evaluating the RF outside its intended phase regime. We caveat that `ndethist` at the retrieved alert is an imperfect phase indicator, and a definitive test would require rescoring archival early-phase alerts.

Table 7. Calibration comparison across brokers and classifiers.

Classifier	ECE	95% CI	ECE (held-out TS)	T	ROC-AUC
ALeRCE LC	0.259	[0.237, 0.280]	0.015	0.361	···
Fink RF[a]	0.464	[0.437, 0.490]	0.353	$\sim$10	0.502
Fink SNN	0.304	[0.278, 0.332]	0.229	$\sim$10	0.525

[a] `rf_snia_vs_nonia` is documented by Fink as targeting *rising*, early-phase SN Ia candidates, not general-purpose SN Ia posteriors (Section 3.3); its row should not be read as an equal-footing comparison. We note, however, that restricting to early-phase proxy subsets does not improve its metrics (Table 5).

NOTE—ALeRCE post-TS ECE is the 5-fold CV held-out value ($T = 0.361 \pm 0.011$ across folds); the same-set oracle fit gives 0.031. $T \sim 10$: optimiser at upper bound. ROC-AUC is defined for the binary Fink scores only; 0.5 = no ranking signal.

5.7. *Robustness to Bin Construction*

Section 4.1.3 motivates recomputing each headline ECE with equal-mass (adaptive) bins as a check against bin-construction artefacts, since Fink's RF classifier has the most skewed score distribution in this study (90% of mass in the first of 10 equal-width bins). Table 6 shows the comparison. The relative change from equal-width to equal-mass binning is at most 1.0% (Fink RF and SNN), and ALeRCE's aggregate ECE is unchanged to the precision reported. The bootstrap CIs of the two binning schemes overlap substantially for every classifier. We conclude the miscalibration estimates in this paper reflect genuine classifier behaviour rather than a bin-width artefact.

5.8. *Cross-Broker Comparison*

Table 7 summarises calibration results across all three classifiers. ALeRCE's light curve classifier, despite the systematic underconfidence described above, retains strong discriminative structure, and its miscalibration is correctable: held-out temperature scaling reduces ECE to 0.015 (5-fold CV; 0.024 on the independent 60/40 split), below the 0.05 practical reference level. The Fink binary scores, on this sample, fail at the more fundamental level of ranking (AUC $\approx$ 0.5; Table 5), which no post-hoc monotone recalibration can repair.

These findings establish that miscalibration is not uniform across production astronomical brokers, and — importantly — that its *repairability* differs in kind, not just degree. ALeRCE's problem is a distorted mapping from evidence to probability atop a sound ranking; a one-parameter correction fixes it out of sample. On this BTS-matched sample, the Fink binary scores carry no measurable ranking signal for general SNIa identification, so no output-side correction can make them useful posteriors; whether this reflects domain shift from their training distribution or evaluation outside their intended operating point (partially disfavoured by the early-phase subsets) cannot be resolved with the data collected here.

6. DISCUSSION

6.1. *Systematic Underconfidence as a Structural Property*

The dominant finding of this study is that ALeRCE's light curve classifier exhibits systematic underconfidence rather than the overconfidence commonly reported for deep learning classifiers (Guo et al. 2017; Minderer et al. 2021). This distinction matters for operational astronomy. Overconfident classifiers produce dangerous silent failures — high-confidence wrong predictions that are acted upon without human review. Underconfident classifiers produce unnecessary deferrals — objects the classifier effectively discriminates but assigns low confidence, causing them to be flagged for spectroscopic follow-up that is not scientifically required.

In the resource-constrained regime of LSST-era astronomy, where spectroscopic follow-up capacity is approximately 10^5 observations per year against 10^7 nightly alerts, unnecessary deferrals carry a direct opportunity cost. Every spectrum allocated to an object the classifier could have confidently classified correctly is a spectrum not allocated to a genuinely ambiguous object or a rare high-priority target. Quantifying and correcting underconfidence is therefore not merely a technical calibration exercise but a resource allocation problem.

6.2. *Why Random Forest Suppresses Confidence*

The underconfidence we observe is consistent with a known property of Random Forest classifiers: probability estimates computed as fractions of tree votes are compressed away from the extremes, particularly under class imbalance (Niculescu-Mizil & Caruana 2005; Zadrozny & Elkan 2002). One component of this picture is directly demonstrated in our sample (Section 5.3): the classifier's output range is observably truncated — no object in 1,576 receives top-1 confidence above 0.88, and among correctly classified SNIa the maximum is 0.72 against an achieved accuracy of 0.83. Whatever its ultimate cause in ALeRCE's training configuration (which we cannot inspect), this compression mechanically guarantees a confidence deficit for any class classified more accurately than the output ceiling.

This accounts for the inverse relationship between accuracy and the true-class-conditioned confidence deficit (definition 1 of Section 4.1.1): SNIa and SNIbc are classified with accuracies above 0.83 — beyond the reachable confidence range — while SNII, at 0.495 accuracy, sits comfortably within it. We emphasise that this inversion is specific to the class-conditional view; under the standard one-vs-rest definition, miscalibration is more uniform across classes (Table 3).

The practical implication is that RF-based astronomical classifiers should not be used for confidence-based deferral decisions without post-hoc calibration. The raw probability outputs do not represent reliable confidence estimates, even when classification accuracy is high.

6.3. *Temperature Scaling: Efficacy and Limits*

Held-out temperature scaling ($T = 0.361 \pm 0.011$ across CV folds) reduces ALeRCE's ECE from 0.259 to 0.015 on held-out folds — a 94% improvement with a single scalar parameter requiring no model retraining, confirmed at 0.024 on an independent 60/40 split. This result is encouraging for operational deployment: existing broker infrastructure could be augmented with a post-hoc calibration layer at negligible computational cost, and the close agreement between held-out and oracle (0.031) fits indicates the correction is stable rather than overfit.

However, the per-class analysis reveals an important limitation. Global temperature scaling worsens SNII calibration from 0.099 to 0.189, because SNII was already well-calibrated before scaling. A single T cannot simultaneously optimise calibration for classes with heterogeneous miscalibration structure.

Two approaches could address this. First, class-specific temperature parameters — fitting a separate T_k for each class k — would allow independent correction of each class's calibration curve. Second, more expressive post-hoc methods such as isotonic regression (Zadrozny & Elkan 2002) or matrix scaling (Kull et al. 2019) could model class interactions that scalar temperature cannot capture. We leave systematic comparison of these methods to future work, noting that the simple temperature scaling result already achieves practically useful calibration for the majority of classes.

We caution that this class-level picture is most secure for SNIa and SNIbc ($n \geq 204$, relative CI width $\lesssim 25\%$). SLSN's pre-scaling ECE alone carries a 95% CI half as wide as its point estimate ($n = 55$; Table 3), so its specific pre/post-scaling trajectory should be read as directionally suggestive rather than a precise per-class calibration measurement. A larger SLSN sample would be needed before treating it as equally established as the SNIa/SNIbc/SNII pattern.

6.4. *Fink: Ranking Failure, Not Merely Miscalibration*

The Fink results are a qualitatively different problem from ALeRCE's underconfidence, and the ranking metrics of Section 5.6 let us state it precisely. With ROC-AUC = 0.502 (RF) and 0.525 (SNN) on this sample, the scores carry essentially no information about which objects are SNIa. This reframes the temperature-scaling failure: the optimiser hitting its bound is a symptom, not the disease. Because monotone recalibration preserves ranking, *no* output-side correction — temperature, Platt, isotonic, beta — can convert an uninformative score into a useful posterior. Our isotonic experiment makes the point concretely: it achieves ECE ≈ 0.01 by collapsing predictions toward the base rate, a "calibrated" output with no per-object content. Any calibration audit that reported ECE alone would mistake this for success.

Why the scores are uninformative here remains open. Candidate explanations are domain shift between Fink's training distribution and the BTS sample, evaluation at alert epochs unlike those the models were tuned for, or limitations of the binary formulations themselves. The early-phase proxy subsets (Table 5) disfavour — but, given the roughness of the `ndethist` proxy, do not exclude — the phase-mismatch explanation for the RF. A definitive attribution would require rescoring archival alerts at controlled phases, which we leave to future work. We stress the scope of the claim accordingly: these scores are unsuitable as general-purpose SNIa posteriors *on a BTS-like bright sample queried at the most recent alert*; we make no claim about their performance in Fink's internal early-alert filtering context.

SuperNNova's marginally higher AUC (0.525) is consistent with the weak rise of its reliability curve, but the difference from chance is too small to support the architectural conclusions (Möller & de Boissière 2020; Villar et al. 2019) we might otherwise be tempted to draw.

6.5. *Implications for Deferral-Aware Classification*

This calibration study motivates the deferral-aware classification framework that constitutes the broader research programme of which this paper is the first component. Learning-to-defer frameworks (Madras et al. 2018; Mozannar & Sontag 2020) require reliable confidence estimates as input to the deferral policy: an object should be deferred when classifier uncertainty genuinely reflects the difficulty of the classification, not when a well-discriminating classifier mechanically suppresses its own signal.

Our results suggest that ALeRCE, post-calibration, is a promising starting point for deferral-aware frameworks (ECE = 0.015 under held-out temperature scaling), pending validation on fainter, LSST-like samples. Fink's binary scores, in their current form and on this sample, are not. This finding directly informs broker selection for downstream deferral framework development.

The taxonomy gap identified in Section 5.4 represents a separate but related challenge. Deferral frameworks require that all scientifically relevant classes appear in the classifier's output space, or that objects outside it be caught by a novelty channel. A deployed classifier without a TDE class will not flag TDE candidates as uncertain on the basis of its probability vector alone — it will instead assign them to the nearest available class, often with moderate-to-high confidence, producing the silent failure mode that deferral is designed to prevent. The recently developed TDE-aware extension (Pavez-Herrera et al. 2025) addresses the taxonomy side of this for TDEs specifically; the general lesson — that a fixed-taxonomy classifier needs either complete coverage or explicit out-of-distribution detection before it can support follow-up allocation — is what carries forward to the deferral framework.

6.6. *Limitations*

Several limitations of this study should be acknowledged. First, we evaluate calibration on a single broker snapshot; ALeRCE and Fink update their classifiers periodically, and our ALeRCE sample spans two distinct production classifier versions active during our data collection period (Section 5.2). We tested rather than assumed that pooling these versions is valid: ECE and the fitted temperature agree closely between them (Table 2), so this limitation does not appear to affect our conclusions for ALeRCE. We did not perform the equivalent check for Fink, and the underlying model could in principle have changed during our collection window; this remains an open caveat for the Fink results specifically. Second, the BTS magnitude limit restricts our sample to bright, low-redshift transients; calibration properties may differ for fainter populations that will dominate LSST alert streams. Third, while our primary temperature-scaling results now use held-out evaluation (5-fold CV and an independent 60/40 split; Section 5.5), deployment would still require refitting T on data from the operating classifier version and monitoring for drift as models are updated.

The second point deserves emphasis because it directly bears on this paper's own motivation: our LSST-oriented recommendations in Sections 6.1, 6.5, and the Conclusions are grounded in a bright, spectroscopically-confirmed sample ($m < 18.5$), while LSST's alert stream will be dominated by fainter, noisier, more ambiguous detections outside the regime we test. As Section 3.4 states, our results are an upper bound on broker calibration performance. Every claim in this paper about LSST-era deployment is therefore an extrapolation from the bright regime, not a direct measurement of the faint regime it is meant to inform; validating that the same qualitative patterns (confidence-compressed underconfidence for ALeRCE, absent ranking signal for the Fink binary scores) persist near the survey's magnitude limit is a necessary next step before any of the recommendations here should be treated as LSST-ready.

Finally, ECE as defined in Equation 2 is known to be sensitive to bin width and the distribution of samples across bins (Nixon et al. 2019), which matters most for highly skewed confidence distributions such as Fink RF's (Section 5.7). We directly tested this rather than leaving it as a caveat: recomputing every headline ECE with equal-mass (adaptive) bins (Roelofs et al. 2022) changes the point estimate by at most 1.0% (Table 6), with substantially overlapping bootstrap CIs. Bin construction is therefore not driving any conclusion in this paper.

7. CONCLUSIONS

We have presented a systematic calibration audit of production astronomical transient classifiers, evaluating ALeRCE's light curve classifier and two Fink classifiers against 1,606 spectroscopically confirmed ZTF transients from the Bright Transient Survey. Our results demonstrate that, in this bright spectroscopic sample, broker probability outputs are not automatically reliable confidence estimates: miscalibration is systematic, broker-dependent, and consequential for confidence-based follow-up prioritisation.

ALeRCE's classifier exhibits systematic underconfidence (ECE = 0.259) with a directly observed cause — its confidence range is compressed well below the accuracy of its best-classified types — consistent with Random Forest vote-averaging under class imbalance. This miscalibration is correctable out of sample: temperature scaling fitted under held-out validation ($T = 0.361 \pm 0.011$) reduces ECE by 94% to 0.015, requiring no model retraining. The correction is not uniform across classes — SNII, well-calibrated in the class-conditional sense before scaling, worsens under a global temperature — motivating class-aware calibration schemes for future work. TDE is absent from the *deployed* light curve classifier's output taxonomy (a TDE-aware version exists but was not serving the stream; Pavez-Herrera et al. 2025), an operational coverage gap worth verifying before LSST first light. On this sample, Fink's binary SNIa scores fail at the level of ranking (ROC-AUC $\approx$ 0.5) rather than calibration alone; no monotone post-hoc correction can render them informative, and phase-restricted proxy subsets do not change this picture.

These findings suggest three priorities for LSST-era broker development. First, calibration metrics — ECE, reliability diagrams, and per-class confidence intervals — should be reported alongside accuracy *and ranking metrics* for production classifiers as standard practice; our Fink isotonic experiment shows that ECE alone can be trivially satisfied by an uninformative score. Second, post-hoc calibration layers should be integrated into broker pipelines where the underlying ranking supports them; the negligible computational cost of temperature scaling is justified by the substantial improvement in probability reliability. Third, taxonomy completeness must be verified against LSST science priorities before deployment. The calibrated ALeRCE outputs demonstrated here are a promising starting point for the deferral-aware classification framework we develop in subsequent work — pending validation on held-out calibration splits from the operating classifier version and on fainter, LSST-like samples — where reliable confidence estimates are prerequisite to learning when to involve human astronomers in follow-up decisions.

We flag one caveat that qualifies all three priorities: they are derived from a bright, spectroscopically-confirmed sample ($m < 18.5$), an upper bound on broker performance (Section 3.4), while LSST's alert stream will be dominated by fainter, higher-redshift, noisier detections we do not test here. These priorities should be understood as hypotheses motivated by the bright regime, to be confirmed on fainter populations before they inform LSST-era broker design decisions.

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